Mustapha Muhsin

FULL-STACK & AI ENGINEER

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Full-stack engineer who builds clear interfaces and reliable backend systems. Experience spans React and Next.js, API design, PostgreSQL-backed applications, cloud infrastructure, asynchronous workflows, and AI-powered software. Comfortable moving between TypeScript, Python, frontend development, and deployment to deliver complete products.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++

Frontend: React, Next.js, Tailwind CSS, responsive interfaces

Backend & Databases: FastAPI, Node.js, Express, PostgreSQL, MySQL, DynamoDB, Redis, Drizzle ORM, Prisma, Auth.js, RESTful APIs, Server-Sent Events (SSE)

AWS & Cloud Infrastructure: AWS (Lambda, ECS, Fargate, Step Functions, DynamoDB, S3, IAM, API Gateway, Cognito, Bedrock), Railway, Docker, Vercel

AI Engineering: RAG Pipelines, LLM Integrations, Tool Calling, LangChain, LangGraph

Developer Tools: Git, Docker, Cursor, Claude Code, GitHub Actions, AWS CLI

EXPERIENCE

White Label Resell — Remote - Los Angeles, CA

AI Backend Engineer — *January 2025 - Present*

- Developed user-facing and backend software across web applications, AI software, and cloud systems.
- Worked through design, implementation, testing, and deployment in a remote engineering team.

National Centre for AI & Robotics (NCAIR) — NITDA — Abuja, Nigeria

AI & Embedded Systems Intern — *February 2022 - July 2022*

- Built the hardware prototype for Elonosia AI (an NCAIR-incubated startup for ML-based malaria detection) — designing the automated microscopy capture device with ESP32/ESP8266, Altium PCBs, and 3D-printed enclosures to bridge the model to clinical use.
- Contributed to training and validation of TensorFlow CNN models for blood smear malaria detection.
- Delivered hands-on STEM workshops for children through NITDA's SB4Kids program — using 3D-printed Otto robots and microcontrollers to introduce IoT, robotics, and programming.

SELECTED PROJECTS

RAG Chatbot API | GitHub

- Built a document question-answering API with FastAPI, PostgreSQL, Pinecone, and AWS S3. Celery and Redis process document ingestion in the background; answers cite retrieved source material.
- Added Docker setup, database migrations, and automated checks to make the service easier to run and maintain.

Facial Recognition Attendance API | GitHub

- Developed employee registration, face-based check-in, and attendance history endpoints using FastAPI, OpenCV, dlib, PostgreSQL, and JWT authentication.
- Protected registration with admin access and stored attendance events for later lookup and filtering.

IoT Solar Tracking System | Final-Year Project

- Designed and built an ESP8266-based solar tracker with ThingSpeak remote monitoring and control.
- Integrated the controller, sensors, and mechanical movement into a working final-year prototype.

EDUCATION

Federal University of Technology Minna

B.Eng., Electrical and Electronics Engineering